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Fair Recourse with KANs

Literature Review: Status and Grouping

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Outline

1. **Problem and scope**;
2. **State of the review**: 28 papers logged;
3. **Review structure**: G1–G4;
4. **The four groups**, paper by paper;
5. **Intersections and gap**;
6. **Next steps and decisions**.

Problem and scope

Recourse (Karimi et al. 2021)

$$a^\star(\mathbf{x}) = \arg\min_{a \in \mathcal{A}(\mathbf{x})} c(a; \mathbf{x}) \quad \text{s.t.} \quad h\big(\mathbf{x}^{\mathrm{SCF}}(a; \mathbf{x})\big) = 1$$

$\mathcal{A}(\mathbf{x})$: actionable interventions · c: cost · $\mathbf{x}^{\mathrm{SCF}}$: structural counterfactual of $\mathbf{x}$ under a · h: classifier

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Group fairness of recourse (von Kügelgen et al. 2022)

$\mathbb{E}\big[\, c(a^\star; \mathbf{x}) \mid h(\mathbf{x}) = 0,\ S = s \,\big]$ is equal for every protected group s

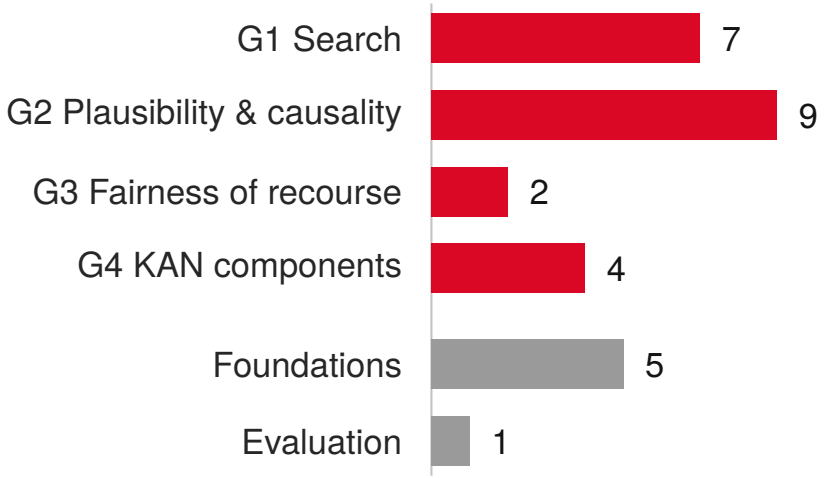
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Thesis goal

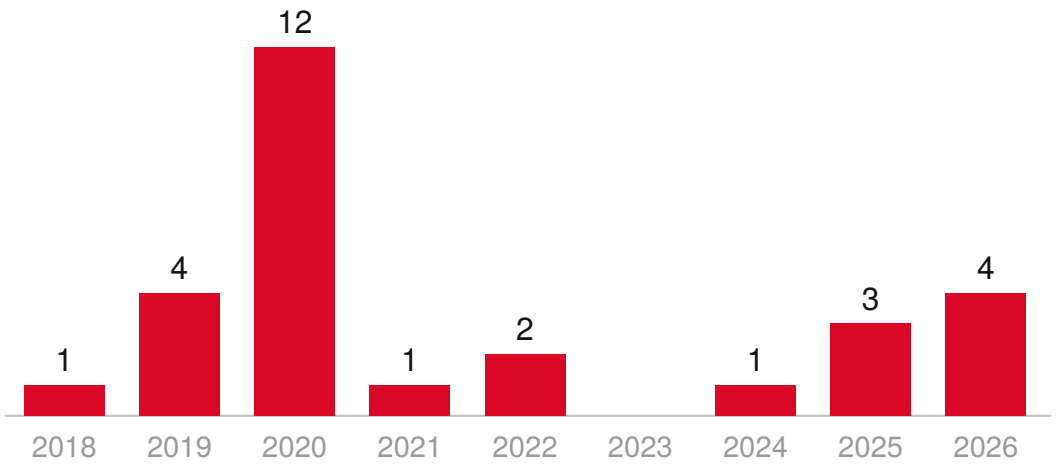
A recourse method that satisfies both, with KANs as one or more of its components: the classifier h, the causal model, or the generator of $a^\star$.

State of the review: 28 papers logged

By group

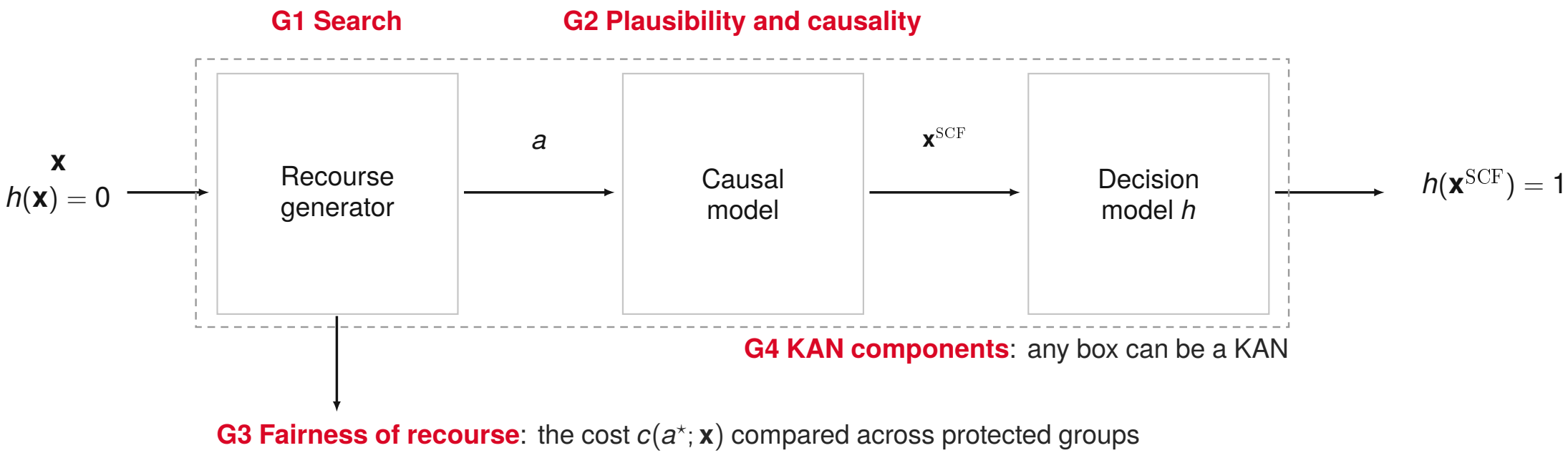


By publication year



16 of 28 papers are from 2019–2020 · 5 more fairness-of-recourse papers identified, not yet logged · no datasets logged yet

Review structure: one group per part of the pipeline



Not tied to one part: **Foundations** (surveys, position papers) and **Evaluation** (datasets, metrics, tools)

Groups, questions and papers

Group	Question it answers	Papers	Chapter 3
G1 Search	How is a valid counterfactual found, with which model access?	7	§3.2
G2 Plausibility and causality	Does it stay on the data and respect causal structure?	9	§3.3
G3 Fairness of recourse	Is the cost of recourse equal across groups and individuals?	2	§3.4
G4 KAN components	Which components have been built with KANs?	4	§3.5
Foundations	Surveys and position papers	5	Ch. 2, §3 intro
Evaluation	Datasets, metrics, tools	1	each section, Ch. 4
Total		28	

Papers logged on 29 Oct 2026 · §3.1 review method, §3.6 synthesis and gaps

G1. Counterfactual search

Method	Year	Search strategy	Model access
Actionable Recourse (Ustun et al.)	2019	Integer programming	Linear
Diverse coherent explanations (Russell)	2019	Mixed-integer programming	Linear
CLEAR (White & d'Avila Garcez)	2019	Local regression surrogate	Black-box
MACE (Karimi et al.)	2020	SMT solving	Model as logic formula
DiCE (Mothilal et al.)	2020	Gradient descent with a diversity term	Differentiable
MOC (Dandl et al.)	2020	Multi-objective evolutionary (NSGA-II)	Black-box
CERTIFAI (Sharma et al.)	2020	Genetic algorithm	Black-box

Exact solvers need the model in closed form; evolutionary and surrogate methods only query it.
Not in the logged papers: encoding spline-based models such as KANs for IP, MIP or SMT solvers.

G2. Plausibility and causality

Method	Year	Mechanism	Causal input
Data manifold			
REVISE (Joshi et al.)	2019	Search in a VAE latent space	None
Prototypes (Van Looveren & Klaise)	2020	Class prototypes guide the search	None
FACE (Poyiadzi et al.)	2020	Shortest paths on a density-weighted graph	None
DACE (Kanamori et al.)	2020	MILP with a Mahalanobis and outlier-aware cost	None
CRUDS (Downs et al.)	2020	Disentangled subspaces of a conditional VAE	None
Causal structure			
Prophit (Lash et al.)	2018	Causal inverse classification, continuous treatments	Treatment model
Mahajan et al.	2020	VAE with a causal-constraint loss	Constraints or SCM
Karimi et al.	2020	Probabilistic recourse under uncertainty	Causal graph
Karimi et al.	2021	Recourse as minimal interventions	Full SCM

Data-manifold methods use no causal knowledge; causal methods range from known constraints to a full SCM.

G3. Fairness of recourse

Paper	Year	Fairness notion	Status
CERTIFAI (Sharma et al.)	2020	Group burden: mean counterfactual distance per group	Logged
von Kügelgen et al.	2022	Group and individual fairness of causal recourse	Logged
Fair Recourse for All (Ezzeddine et al.)	2026	Individual and group fairness enforced in generation	Logged
Equalizing Recourse (Gupta et al.)	2019	Group gap in distance to the boundary	Seed
Ehyaei et al.	2023	Robustness and individual fairness of causal recourse	Seed
Kavouras et al.	2023	Subgroup audits with counterfactuals	Seed
Bell et al.	2024	Substantive equality of opportunity	Seed
Barrainkua et al.	2026	Worst-case social burden	Seed

Logged: read and recorded · Seed: identified, not yet logged
The notions differ in what they compare (distance, cost, burden) and in whether they need an SCM.

G4. KAN components

Paper	Year	KAN role	Relation to the thesis
Priyadarshini & Gago-Masague	2025	Model in adversarial fair-representation learning	Fairness with KANs; no recourse
Almodóvar et al. (KaCGM)	2026	Causal generative model	KAN as causal model
Valencia et al.	2026	Counterfactual explanations for an RL agent	KANs and CEs; RL, not tabular
Hoang et al.	2026	On-chip online learning	Out of scope

Surveys (under Foundations): Somvanshi et al. 2025, ACM CSUR · Noorizadegan et al. 2025, Comput. Sci. Rev.
Limitation reported for KaCGM: low-complexity functional decompositions trade accuracy for auditability.

Intersections found so far

Intersection	Logged papers	Count
Recourse × causal structure	Lash et al. 2018; Mahajan et al. 2020; Karimi et al. 2020, 2021; von Kügelgen et al. 2022	5
Recourse × fairness	Sharma et al. 2020; von Kügelgen et al. 2022; Ezzeddine et al. 2026	3
CE × KANs	Valencia et al. 2026 (RL agents)	1
Causal model × KANs	Almodóvar et al. 2026	1
Fairness × KANs	Priyadarshini & Gago-Masague 2025 (representations)	1
Fair recourse × KANs	–	**0**

Counts from the 28 papers logged on 29 Oct 2026, before the systematic search (S1–S4, planned for 2–8 Nov).

Next steps and decisions

1. Log the 5 remaining fairness-of-recourse seeds — 1 Nov
2. Run S1–S4 in all sources; log counts — 8 Nov
3. Log datasets, metrics and benchmarks (CARLA) — 15 Nov
4. Full Chapter 3 draft — 20 Dec

Experiments

Are preliminary results expected in January (e.g. KAN vs MLP on Adult)?

Scope

Include RL-based CE work (Valencia et al. 2026), or only tabular classification?

Delivery date

What is the exact January deadline? The plan assumes ~22 Jan.